Motion & Forces

Physics concept review

Study snapshot

Use this guide to connect motion graphs with the forces that change motion.

Key ideas

- Distance is total path length; displacement includes direction.
- Average speed equals distance divided by elapsed time.
- Acceleration is the rate of change of velocity: a = (v - u) / t.
- A net force changes an object’s momentum and can produce acceleration.
- Draw and label a free-body diagram before resolving forces.

Try the questions on page 2 before checking your work.

Motion & Forces

Quick practice

1. A runner covers 120 m in 20 s. Find the average speed.
2. Velocity changes from 4 m/s to 16 m/s in 3 s. Find acceleration.
3. Name the direction of friction for a box sliding right.
4. What does a horizontal position-time graph show?

Answer notes

1) 6 m/s
2) 4 m/s^2
3) To the left, opposing the sliding.
4) The object is at rest.

Review the method, then retry any question you missed.